TAI, Wing-Kuen (William)

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SUMMARY

Computer Engineering graduate from HKUST focused on backend systems and AI application pipelines. Experienced in designing asynchronous REST APIs (FastAPI), deploying Docker microservices, and implementing LLM tool-calling and OCR workflows across web and mobile platforms.

EDUCATION

The Hong Kong University of Science and Technology (HKUST)

Sep 2023 – Jun 2026

Bachelor of Engineering in Computer Engineering

• **Relevant Coursework:** Object-Oriented Programming & Data Structures, Operating Systems, Database Management Systems, Computer Networks, Embedded Systems (STM32/Arduino MCU & C/C++)

• **Practical Engineering Training:** Electronic Design & 2-Layer PCB Fabrication (USTM30), Linux System & Network Administration (USTM17), Cloud Services in Microsoft Azure (USTM25)

HKU SPACE Community College

Sep 2021 – Aug 2023

Associate of Engineering in Computer Science

TECHNICAL SKILLS

Programming Languages: Python, Kotlin, TypeScript, JavaScript, SQL, C/C++, Java, Dart, Bash

AI, Vision & Agent Orchestration: LLM Function Calling, Model Context Protocol (MCP), Vision-Language Models (Qwen-2.5-VL), PaddleOCR-VL-1.5 (PP-Structure), YOLO26l, OpenVINO CPU Quantization (INT4/INT8), OpenCV, NumPy, Hugging Face APIs, Prompt & JSON Schema Engineering

Backend, Cloud & Linux Infrastructure: Python (FastAPI), RESTful API Design, Docker, Docker Compose, Linux (Ubuntu, CentOS, systemd, OpenLDAP, SSSD, NFS, firewalld/nftables), Microsoft Azure (Azure SQL PaaS, Key Vault RBAC)

Databases & Storage: MongoDB Atlas, GridFS (Binary Storage), MySQL, Azure SQL (T-SQL), SQLite / Room, Hive (Flutter Local KV Store)

Web & Mobile: Next.js, React, TypeScript, Kotlin (Jetpack Compose), Flutter, Tailwind CSS

Embedded & Hardware: Embedded C/C++, Microcontrollers (STM32 / Arduino), KiCad (Schematic Capture & 2-Layer PCB Layout, Gerber/Drill), 74HC595 Shift Registers, ADC / PWM / UART Serial, Autodesk Tinkercad

Developer Tools: Git, GitHub Actions (CI/CD Pipelines), Vitest, Mock Service Worker (MSW), Postman

TECHNICAL PROJECTS

AI-Powered Multi-Modal Health Platform

Jun 2025 – May 2026

HKUST Final Year Project (Grade A) | Backend Lead & Systems Developer (4-person team)

- **Built an asynchronous FastAPI backend** on Ubuntu serving 30+ endpoints with bcrypt authentication, daily check-in idempotency guards, and midnight cron resets via APScheduler
- **Created a two-phase Scan-and-Confirm API** (/upload-image and /confirm-foodlog partial updates via Pydantic), plus endpoints for Nutri-Score calculation, nutrition trend charts, and leaderboards
- **Integrated Qwen-2.5-VL-72B** (AsyncOpenAI) with strict JSON schema validation, regex sanitization, fallback objects, and HTTP 503 backoff to prevent API rate-limit errors

- **Reduced prompt payloads** from thousands of tokens to under 100 by aggregating 7-day MongoDB records into compact daily summaries, stripping null fields, and enforcing per-endpoint token budgets
- **Automated a 6-stage in-memory OCR pipeline** in Docker Compose combining local YOLO26l detection (96.6% accuracy, 98.3% F1, under 200ms crop), PaddleOCR-VL-1.5 table parsing, and MongoDB GridFS streaming, eliminating intermediate disk I/O
- **Tuned PaddleOCR-VL-1.5 with OpenVINO** (INT4/INT8 quantization), cutting model size by 56% (from 1,477MB to 644MB) and eliminating JIT cold-start stalls with tensor warm-up to reduce CPU inference latency from 15.5s to 11.5s per image

Smart Trip Planner (Agentic Travel App) Sep – Nov 2025

Mobile & Cloud Integration (HKUST COMP4521) | Flutter, MVVM, Gemini Tool Calling, MongoDB Atlas

- **Implemented a multi-turn LLM tool-calling loop** with Gemini 2.5 Flash Lite using MVVM architecture with Provider, intercepting dynamic user queries across external flight and hotel APIs to synthesize structured multi-day itineraries
- **Integrated 7 Amadeus and Google Places REST endpoints** featuring automated OAuth2 token refresh, free-text to IATA airport code resolution, two-stage batch hotel pricing, and parallel media fetching with timeout guards
- **Engineered a 2-tier local caching layer and cloud persistence** combining in-memory and Hive NoSQL stores with Stale-While-Revalidate refresh, backed by MongoDB Atlas with compound user and date indexes for sub-second retrieval

AI Developer Toolchain, MCP Servers & Multi-Agent Task Pipeline Jan – Sep 2026

Personal Project | Model Context Protocol (MCP), Docker, Chrome DevTools MCP, Python, Git

- **Deployed containerized Model Context Protocol (MCP) servers** using Docker to securely bridge local AI coding assistants with GitHub APIs and local filesystem operations
- **Designed a 3-stage multi-agent automation pipeline** coordinating specialized discovery, rubric-scoring, and Chrome DevTools MCP execution agents to automate web DOM extraction, document generation, and React form state synchronization

Web Crawler, Positional Inverted Index & Search Engine Spring 2026

Information Retrieval Project (HKUST COMP4321) | Python, SQLite, BeautifulSoup, NLTK

- **Built a BFS web crawler and 5-table SQLite positional inverted index** with HTTP Last-Modified incremental cache invalidation, NLTK Porter stemming, and word-position tracking for exact-phrase search and title-boosted cosine similarity ranking

TimeLock Anti-Addiction Android Application Aug 2026

Personal Project | Kotlin, Jetpack Compose, Room / SQLite, Material 3

- **Built a state-machine and persistent countdown** using StateFlow and a Room heartbeat, preventing background coroutine freezes and timing drift across app switches
- **Implemented global cooldown and 60-second emergency pass** to curb cross-app usage, enforcing single-use tokens to prevent cooldown bypass
- **Resolved Android edge cases and UI consistency**, including Xiaomi background event suppression, WindowMetrics orientation clamping, and Material 3 theme injection for WindowManager overlays

React Testing & CI Pipeline Showcase Sep 2026

Personal Project | Vitest, React Testing Library, MSW, GitHub Actions

- **Wrote integration test suites** using Vitest and React Testing Library for authentication pages, using Mock Service Worker (MSW) to simulate 401, 404, and 500 network error states
- **Configured GitHub Actions CI workflow** with npm caching to run automated test suites and collect coverage artifacts on pull requests

Interactive Portfolio Web Application Jun 2026

Personal Project | Next.js, React, TypeScript, Tailwind CSS

- **Built responsive portfolio** with Next.js App Router, implementing dynamic video lazy-loading to optimize First Contentful Paint (FCP)

8x8 LED Matrix & Shift-Register Hardware Controller 2024

Embedded Systems & PCB Engineering (HKUST USTM30) | Embedded C/C++, KiCad, 74HC595, Tinkercad

- **Simulated and designed 2-layer custom PCB** in Autodesk Tinkercad and KiCad (schematic capture, PCB routing, Gerber and NC Drill generation), completing physical THT/SMD soldering and hardware verification
- **Developed bare-metal C/C++ firmware** driving daisy-chained 74HC595 shift registers, time-multiplexed 8x8 LED dot-matrix animations, a 7-segment display, ADC potentiometer control, PWM piezo buzzer audio, and 4 directional buttons debugged via UART serial

Enterprise Linux Directory, Shared Storage & Azure Cloud Infrastructure 2024

Systems & Cloud Engineering (HKUST USTM17 & USTM25) | CentOS Linux, OpenLDAP, NFS, Azure SQL, Key Vault

- **Configured Linux directory and network storage services** on CentOS 8 by compiling OpenLDAP from source with OpenSSL TLS and crypt support, designing hierarchical LDIF schemas, integrating SSSD client authentication, and configuring NFS shared storage with root_squash and firewalld/nftables rules
- **Provisioned Microsoft Azure PaaS and RBAC security** by deploying Azure SQL Databases with server-level IP firewall whitelisting, resolving Control Plane vs. Data Plane IAM RBAC permissions for Azure Key Vault secrets, and modeling infrastructure hosting costs

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WORK EXPERIENCE

IT Deployment Technician (Contract) | Proseek Technology Sep 2026 – Present

Enterprise Infrastructure Project (Client: UBS, West Kowloon HQ)

- **Deployed enterprise workstations** and dual-monitor displays across UBS trading floors, setting up hardware, power distribution, and cable management to bank workspace standards
- **Audited hardware assets** by checking BIOS configurations and recording serial numbers as part of a 500+ endpoint enterprise rollout

Office Assistant | Hayes Cleveland Limited Jul 2024 – Nov 2024

- **Managed inventory reconciliation** and order fulfillment tracking via Google Sheets for an active e-commerce storefront
- **Coordinated parcel logistics** and handled customer inquiries via email and messaging with same-day turnaround

Store Assistant | Circle K Jun 2023 – Aug 2023

- **Executed POS transactions**, cash reconciliations, and inventory restocking in a high-volume retail branch
- **Handled customer service** and e-commerce pickup fulfillment workflows

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CERTIFICATIONS

- Software Engineering: Modeling Software Systems using UML | HKUST via Coursera Jul 2026
- Gemini Certified University Student | Google for Education Jun 2026
- AI Code Generation for Everyone | HKUST via Coursera Jun 2026
- Search Engines for Web and Enterprise Data | HKUST via Coursera May 2026

LANGUAGES

Cantonese (Native) | English (Fluent) | Mandarin (Fluent)